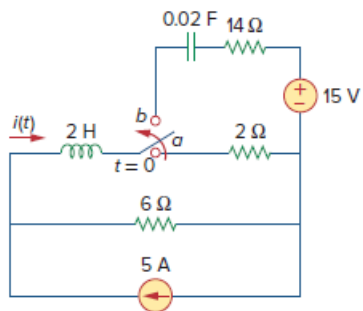


Problem

The switch in the circuit of Fig. 16.61 is moved from position a to b (a make before break switch) at $t = 0$. Determine $i(t)$ for $t > 0$.

Figure 16.61:



Step-by-step solution

Step 1 of 6

Refer to circuit diagram in Figure 16.61 in the textbook.

The switch is at position a for $t < 0$. When $t < 0$, circuit is in steady state. Under steady state, inductor acts as a short circuit.

Draw the circuit for $t < 0$ as shown in Figure 1.

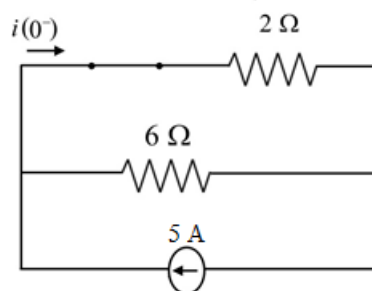


Figure 1

Step 2 of 6

Calculate $i(0^-)$ using current division principle.

$$\begin{aligned} i(0^-) &= \left(\frac{6\ \Omega}{2\ \Omega + 6\ \Omega} \right) (5\ \text{A}) \\ &= \left(\frac{6\ \Omega}{8\ \Omega} \right) (5\ \text{A}) \\ &= (0.75)(5\ \text{A}) \\ &= 3.75\ \text{A} \end{aligned}$$

The current through the inductor doesn't allow sudden change.

So,

$$\begin{aligned} i(0^-) &= i(0^+) \\ &= i(0) \end{aligned}$$

Step 3 of 6

When $t > 0$, the switch is at position b . Draw the s-domain circuit as shown in Figure 2.

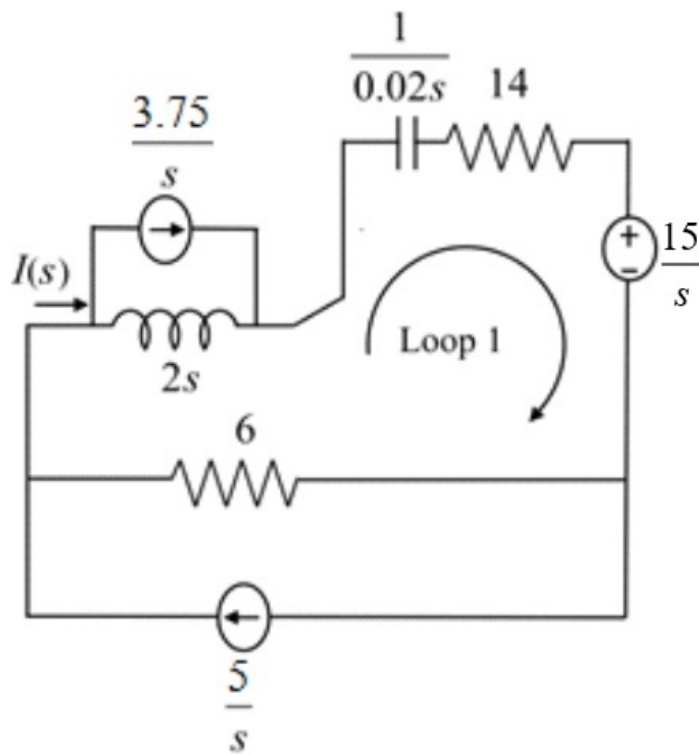


Figure 2

Step 4 of 6

The initial voltage of the capacitor is zero.

Apply Kirchhoff's voltage law to the Loop 1 shown in Figure 3.

$$2s \left[I(s) - \frac{3.75}{s} \right] + \left[\frac{1}{0.02s} + 14 \right] I(s) + \frac{15}{s} + 6 \left[I(s) - \frac{5}{s} \right] = 0$$

$$-7.5 + \left[2s + 6 + \frac{1}{0.02s} + 14 \right] I(s) + \frac{15}{s} - \frac{30}{s} = 0$$

$$\left[\frac{0.04s^2 + 0.4s + 1}{0.02s} \right] I(s) = \frac{15 + 7.5s}{s}$$

$$I(s) = \frac{15 + 7.5s}{2s^2 + 20s + 50}$$

$$= \frac{7.5 + 3.75s}{s^2 + 10s + 25}$$

$$= \frac{3.75s + 7.5}{(s + 5)^2}$$

Step 5 of 6

Simplify the equation using partial fraction.

$$\frac{3.75s + 7.5}{(s+5)^2} = \frac{A}{(s+5)} + \frac{B}{(s+5)^2}$$

$$3.75s + 7.5 = A(s+5) + B$$

Compare the s coefficients on both sides.

$$A = 3.75$$

Compare the constants on both sides.

$$7.5 = 5A + B$$

$$7.5 = 5(3.75) + B$$

$$B = -11.25$$

Step 6 of 6

Substitute 3.75 for A , and -11.25 for B in the expression of $I(s)$.

$$I(s) = \frac{3.75}{(s+5)} - \frac{11.25}{(s+5)^2}$$

Apply inverse Laplace transform.

$$i(t) = [3.75e^{-5t} - 11.25te^{-5t}]u(t) \text{ A}$$

Therefore, the expression of $i(t)$ for $t > 0$ is $\boxed{[3.75e^{-5t} - 11.25te^{-5t}]u(t) \text{ A}}$.